

Introduction to Product Design: Beyond Engineering

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Agenda

1. Deconstructing Your Week 1 Experience
2. What Makes a Product?
3. Stakeholders & Constraints
4. The Designer-Engineer

Part 1: Deconstructing the Week 1 & 2 Experience

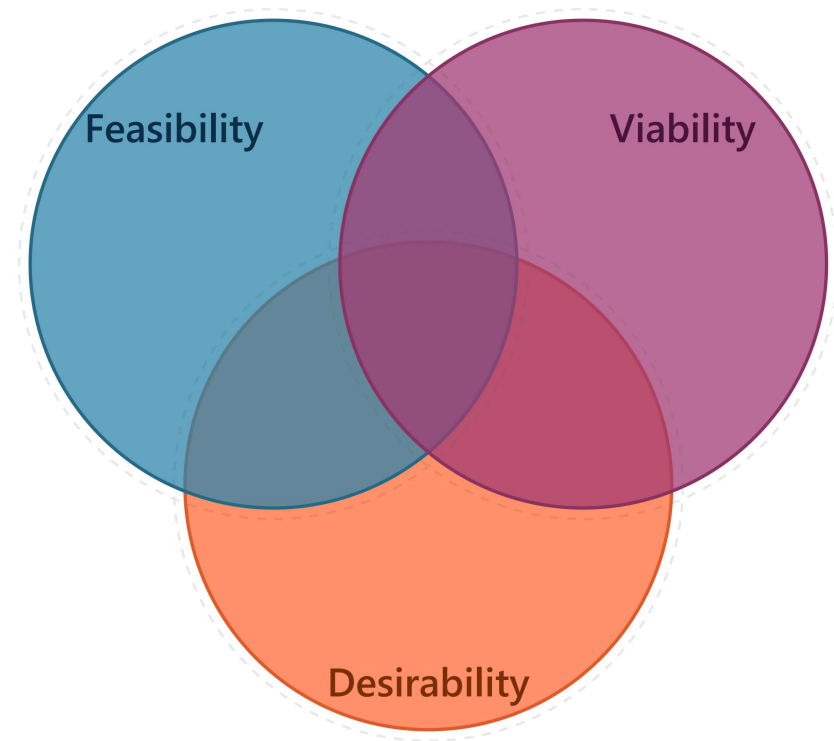
The "It Works" Trap

You built a light that increases visibility.
It turned on. It was bright. The cost was less than €5.

But was it a product?

Three Questions Every Product Must Answer

1. **Feasibility:** Can we build it?
(Your comfort zone: circuits, code, 3D printing)
2. **Viability:** Should we build it?
(Does it make economic/strategic sense?)
3. **Desirability:** Does anyone want it?
(Will people actually use it in -20°C darkness?)



Interactive Activity 1: Reflection (10 min)

In your Week 1 groups, discuss:

- Which of the three (Feasibility, Viability, Desirability) was **hardest** for your team?
- What assumption did you make that turned out to be wrong?
- If you had another week, what would you change?

Be ready to share one insight with the class.

Class Discussion (10 min)

Let's hear from some groups:

- What was the hardest part of the €5 limit?
- Was it the **components** or the **choices**?

Key Takeaway from Week 1

"A working prototype is the beginning of product design, not the end."

Engineering gets you to feasibility.

Product design gets you to **market success**.

Part 2: What Makes a Product?

Is this chocolate cake a product?



Defining "Product"

Product (Ulrich & Eppinger):

"Something sold by an enterprise to its customers."

In our context:

A physical or cyber-physical system that delivers **value** to users within **constraints**.



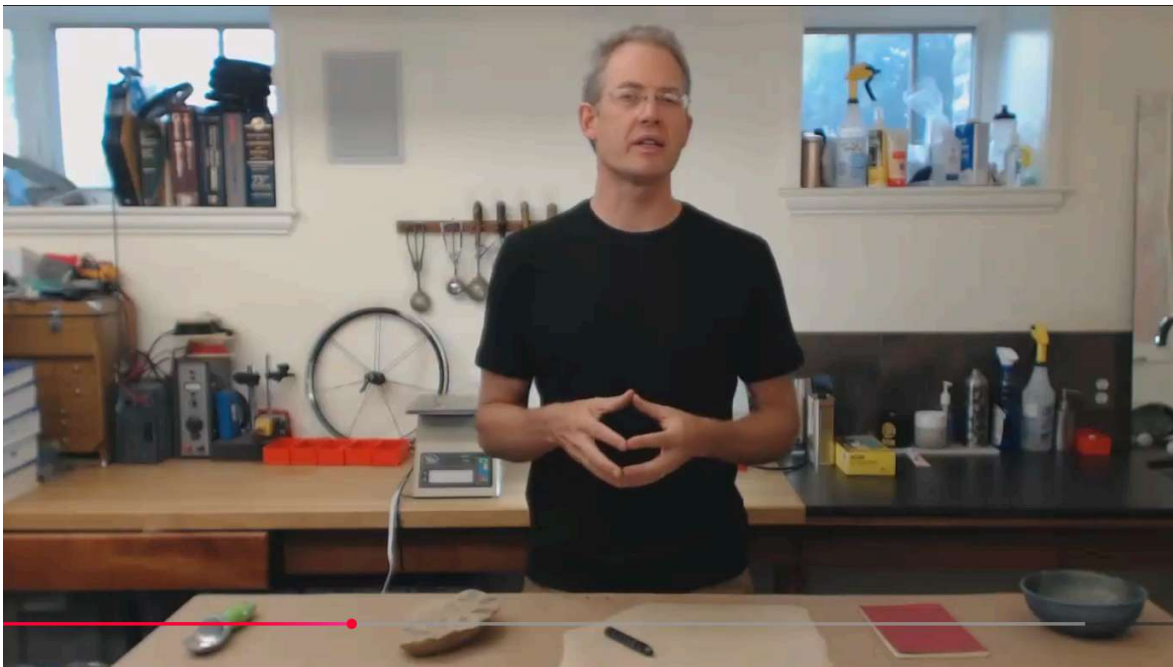
Project vs. Product

Engineering Project	Product
Solves a technical challenge	Solves a user problem
Works in the lab	Works in the real world
One-off prototype	Reproducible at scale
Evaluated by specs	Evaluated by adoption

If your Week 1 challenge was more like a project, now let's move into a product mindset.

What is design?

https://youtu.be/hzNVnN_765U?si=2PXuHiC3gyzl-TFI&t=269 (From 4:05 to 8:08)

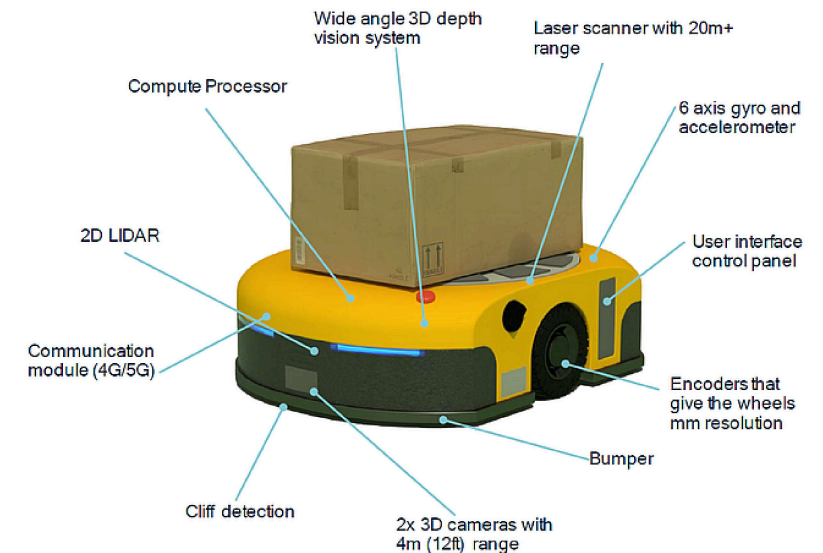


Functions vs. Features

- **Function:** *What the product does* (increase visibility)
- **Feature:** *How it does it* (500-lumen strobe, motion sensor, reflective housing)

Example in Robotics:

- **Function:** Autonomous navigation
- **Features:** LiDAR, SLAM algorithm, obstacle avoidance, 8-hour battery



Product Architecture

How do the **subsystems** fit together?

Example: Your Visibility Light

- Power subsystem (battery, voltage regulation)
- Light subsystem (LED, driver circuit)
- Control subsystem (switch, sensor, microcontroller?)
- Enclosure subsystem (housing, mounting, weatherproofing)

These are all the parts in a car.

What subsystems can you identify in a combustion engine car? And in an electric one?



The "Total Product" Concept

A product is more than hardware:

- Packaging
- Instructions / Documentation
- Charging method
- Maintenance
- End-of-life / Disposal

In robotics: Software updates, calibration tools, spare parts, support

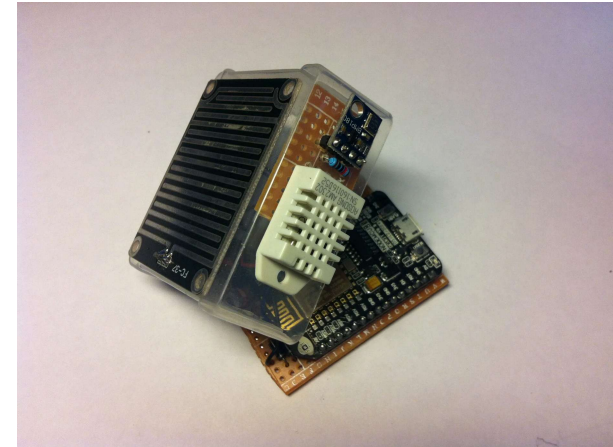
Case Study: DIY vs. Product

DIY IoT Weather Station (Engineering Project)

- Raspberry Pi + sensors
- Works on your desk
- Requires technical knowledge

Netatmo Weather Station (Product)

- Plug-and-play
- Mobile app
- Designed for non-engineers



Interactive Activity 2: Product Teardown (20 min)

In groups, pick one:

- A consumer robot (for instance, Roomba, VEX Aim, Anki Cozmo)
- A wearable device (Fitbit, smartwatch)
- An IoT device (smart thermostat, smart lock)

Identify:

1. Main functions
2. Key features
3. Product architecture (subsystems)

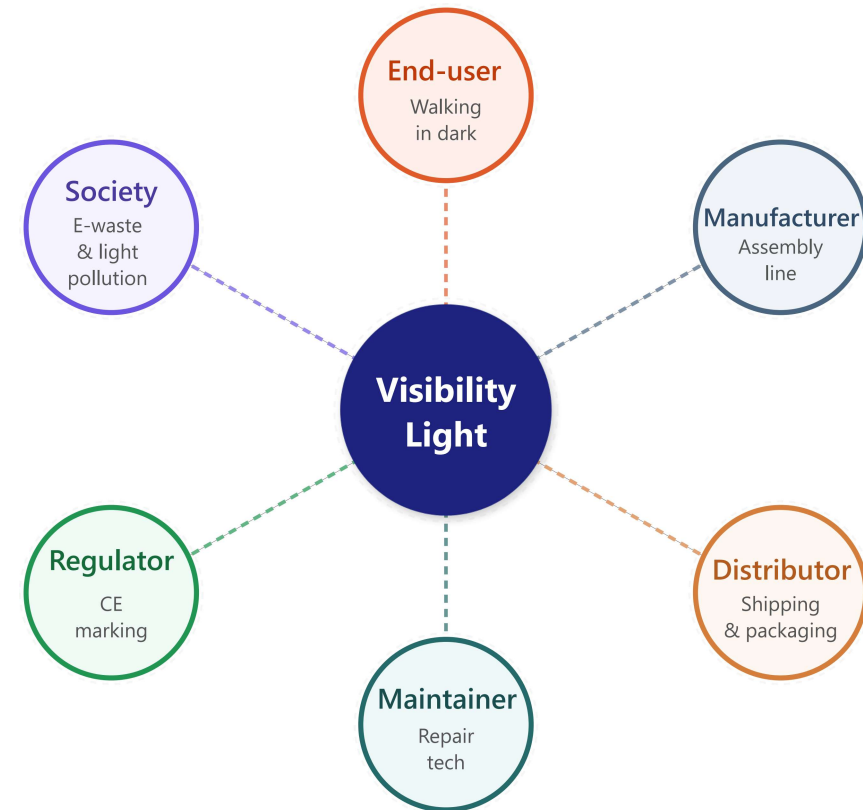
Share one insight with the class.

Part 3: Stakeholders & Constraints

Who Are We Designing For?

Not just the end-user!

- **End-user** (the person walking in the dark)
- **Manufacturer** (assembly line worker)
- **Distributor** (packaging, shipping)
- **Regulator** (CE marking, battery safety)
- **Maintainer** (repair technician)
- **Society** (light pollution, e-waste)



Example: Agricultural Robot

End-user: Farmer

Other stakeholders:

- Veterinarian (animal safety)
- Repair technician (serviceability)
- Neighbor (noise, dust)
- Regulator (machinery safety standards)
- The animals themselves (stress, behavior)



Identifying Constraints

Economic:

- Bill of Materials (BOM) cost
- Manufacturing cost
- Time-to-market

Environmental:

- Operating temperature (-20°C in Finland)
- Moisture, dust, vibration
- Outdoor vs. indoor use

Identifying Constraints (cont.)

Legal / Ethical:

- CE marking (EU)
- Battery regulations (shipping, disposal)
- Light pollution (blinding drivers?)
- Data privacy (if IoT-connected)

Technical:

- Battery life
- Size and weight
- Durability (drop tests, water resistance)



Part 4: The Designer-Engineer

Your Role in Product Design

You are **not just** implementers

You are **decision-makers** who balance:

- Technical performance
- User needs
- Business constraints
- Ethical considerations

The T-Shaped Engineer

Vertical bar (Deep):

Your technical expertise (Electronics, IoT, Robotics, Programming)

Horizontal bar (Broad):

Empathy, communication, business thinking, design thinking, sustainability

Quality Assurance	Project Manager	Software Engineer	Technical Writer	Manager
UI tests	Prioritize tickets	XML	Internal documentation	Interview candidates
Compatibility tests	Collaborate	APIs	External documentation	Mentor
Cross-platform tests	Coordinate release	REST	Product copy	Communicate
Automation tests	Plan features	Postman		Gauge team morale
	User stories	JSON		
		SQL		
		End to end tests		
		Normalize data		
		Design data models		
		Support customers		
		Peer code reviews		
		Git		
		Unit tests		

Trade-offs: The Heart of Design

You **cannot** have:

- The brightest light
- The longest battery life
- The lowest cost
- The smallest size

...all at the same time

Design is about making informed trade-offs

Example Trade-off: Robot Gripper

Option A	Option B
High grip force	Gentle grip
Heavy, expensive	Light, cheap
Slow actuation	Fast actuation
Durable	Fragile

Which do you choose?

Answer: *It depends on the application and stakeholders*

Preparing for Lepaa Visit

What to look for:

- **Pain points** (What's frustrating? Dangerous? Slow?)
- **Workarounds** (What hacks do farmers use?)
- **Unmet needs** (What do they wish existed?)
- **Context** (Environment, tools, workflows)

Not: "What cool robot can we build?"

But: "What problem can we solve?"

Preparing for Lepaa Visit

Bring:

- Observation guide (provided on course platform)
- Camera / phone
- Paper and pen
- Curiosity

Key Takeaways from Week 3

1. "It works" $\neq$ "It's a product"
2. Products serve users within constraints
3. Design is about trade-offs, not perfection
4. You are designer-engineers, not just builders

Thank you!

Next week: The Product Design Process